5

Black Holes

The black hole Gargantua plays a major role in *Interstellar*. Let's look at the basic facts about black holes in this chapter and then focus on Gargantua in the next.

First, a weird claim: *Black holes* are made from warped space and warped time. Nothing else—no matter whatsoever.

Now some explanation.

Ant on a Trampoline: A Black Hole's Warped Space

Imagine you're an ant and you live on a child's trampoline—a sheet of rubber stretched between tall poles. A heavy rock bends the rubber downward, as shown in Figure 5.1. You're a blind ant, so you can't see the poles or the rock or the bent rubber sheet. But you're a smart ant. The rubber sheet is your entire universe, and you suspect it's warped. To